

Symmedians

1 Definition and Properties

Definition. In $\triangle ABC$, the reflection of the A -median about the A -internal angle bisector is called the A -symmedian of $\triangle ABC$. The B -symmedian and the C -symmedian of $\triangle ABC$ are defined in a similar way.

Below are some characterizations¹ of symmedians:

- (1) In $\triangle ABC$ with X on side BC , then AX is the A -symmedian if and only if

$$\frac{XB}{XC} = \frac{AB^2}{AC^2}.$$

- (2) In $\triangle ABC$ with X on side BC , then AX is the A -symmedian if and only if for any point P on AX , we have

$$\frac{d(P, AB)}{d(P, AC)} = \frac{d(X, AB)}{d(X, AC)} = \frac{AB}{AC},$$

where $d(U, VW)$ denotes the distance of point U from VW .

- (3) In $\triangle ABC$, let $ACUV$ and $ABST$ be the squares constructed on the sides which are directed towards the exterior of the triangle. Let X be the circumcentre of $\triangle ATV$. Then, the line AX is the A -symmedian of $\triangle ABC$.
- (4) Let MN be a line parallel to the side BC of $\triangle ABC$, with M on the side AB and N on the side AC . The lines BN and CM meet at point Y . The circumcircles of $\triangle BMY$ and $\triangle CNY$ meet at two distinct points Y and X . Then the line AX is the A -symmedian of $\triangle ABC$.
- (5) The symmedian point K of $\triangle ABC$ is the only point in the plane of ABC which is the centroid of its own pedal triangle.
- (6) Let the tangents at vertices B and C of $\triangle ABC$ to its circumcircle meet at a point X . Then the line AX is the A -symmedian of $\triangle ABC$.
- (7) The A -symmedian is the locus of the midpoints of the antiparallels to BC bounded by the lines AB and AC .
- (8) Suppose X is a point of the circumcircle of $\triangle ABC$, different from the vertex A , such that

$$\frac{XB}{XC} = \frac{AB}{AC}.$$

Then the line AX is the A -symmedian of $\triangle ABC$.

2 Practice Problems

Problem 1. Let ABC be a triangle with $AC = BC$, and P a point inside the triangle such that $\angle PAB = \angle PBC$. If M is the midpoint of AB , then show that $\angle APM + \angle BPC = 180^\circ$.

Problem 2. Let P be a point on the circumcircle of $\triangle ABC$, and let D, E and F be the feet of the perpendiculars from P to BC, CA and AB , respectively. Then P lies on the A -symmedian if and only if $DE = DF$.

Problem 3. Let AM be a median of $\triangle ABC$. Suppose the perpendicular bisectors of AC and AB intersect AM at points B', C' , respectively, and lines BC' and CB' meet at A' . Prove that $\angle A'AB = \angle C'AC$.

Problem 4. Acute-angled triangle ABC is inscribed into circle Ω . Lines tangent to Ω at B and C intersect at P . Points D and E are on AB and AC such that PD and PE are perpendicular to AB and AC respectively. Prove that the orthocentre of $\triangle ADE$ is the midpoint of BC .

Problem 5. Let M be the midpoint of the side AC of an acute-angled triangle ABC with $AB > BC$. Let Ω be the circumcircle of $\triangle ABC$. The tangents to Ω at the points A and C meet at P , and BP and AC intersect at S . Let AD be the altitude of $\triangle ABP$ and ω the circumcircle of $\triangle CSD$. Suppose ω and Ω intersect at $K \neq C$. Prove that $\angle CKM = 90^\circ$.

Problem 6. A point B lies on a chord AC of circle ω . Segments AB and BC are diameters of circles ω_1 and ω_2 centered at O_1 and O_2 respectively. These circles intersect ω for the second time in points D and E respectively. The rays O_1D and O_2E meet in a point F , and the rays AD and CE do in a point G . Prove that the line FG passes through the midpoint of the segment AC .

Problem 7. Let AK and AT be the bisector and the median of an acute-angled triangle ABC with $AC > AB$. The line AT meets the circumcircle of $\triangle ABC$ at point D . Point F is the reflection of K about T . Prove that $2\angle FDA = \angle ABC - \angle ACB$.

¹cf. Sammy Luo and Cosmin Pohoata, Let's Talk About Symmedians!, Mathematical Reflections 4 (2013), 1-11.

Problem 8. A circle, its chord AB and the midpoint W of the minor arc AB are given. Take an arbitrary point C on the major arc AB . The tangent to the circle at C meets the tangents at A and B at points X and Y respectively. Lines WX and WY meet AB at points N and M respectively. Prove that the length of segment NM does not depend on point C .

Problem 9. Construct $\triangle ABC$ given its vertex B , its centroid and the intersection point $L \neq B$ of its circumcircle with the B -symmedian.

Problem 10. Let ABC be an acute-angled, nonisosceles triangle. Altitudes AA' and BB' meet at point H , and the medians of triangle AHB meet at point M . Line CM bisects segment $A'B'$. Find $\angle ACB$.

Problem 11. In $\triangle ABC$, let BD be its internal angle bisector BD with D on BC . The line BD intersects the circumcircle Ω of $\triangle ABC$ at B and E . Circle ω with diameter DE cuts Ω again at F . Prove that BF is a symmedian of $\triangle ABC$.

Problem 12. In $\triangle ABC$, let $\angle BAC = 60^\circ$, AD be its internal angle bisector, and let $\triangle PDQ$ be equilateral with altitude DA . The lines PB and QC meet at point K . Prove that AK is a symmedian of $\triangle ABC$.

Problem 13. In $\triangle ABC$, the tangent line to its circumcircle at point C meets line AB at point D . The tangent lines to the circumcircle of $\triangle ACD$ at points A and C meet at point K . Prove that line DK bisects segment BC .

Problem 14. Consider an acute-angled triangle ABC with $AB < AC$ and let ω be its circumscribed circle. Let t_B and t_C be the tangents to the circle ω at points B and C , respectively, and let L be their intersection. The straight line passing through the point B and parallel to AC intersects t_C in point D . The straight line passing through the point C and parallel to AB intersects t_B in point E . The circumcircle of the triangle BDC intersects AC in T , where T is located between A and C . The circumcircle of the triangle BEC intersects the line AB (or its extension) in S , where B is located between S and A . Prove that ST , AL , and BC are concurrent.

Problem 15. Three distinct points A, B, C are fixed on a line in this order. Let Γ be a circle passing through A and C whose centre does not lie on the line AC . Denote by P the intersection of the tangents to Γ at A and C . Suppose Γ meets the segment PB at Q . Prove that the intersection of the bisector of $\angle AQC$ and the line AC does not depend on the choice of Γ .

Problem 16. In the plane, two circles intersect at A and B , and a common tangent intersects the circles at P and Q . Let the tangents at P and Q to the circumcircle of $\triangle APQ$ intersect at S , and let H be the reflection of B across the line PQ . Prove that the points A, S , and H are collinear.

Problem 17. Triangle ABC is inscribed in circle ω . The tangent lines to ω at B and C meet at T . Point S lies on ray BC such that $AS \perp AT$. Points B_1 and C_1 lie on ray ST (with C_1 in between B_1 and S) such that $B_1T = BT = C_1T$. Prove that $\triangle ABC \sim \triangle AB_1C_1$.

Problem 18. Let ABC be a triangle with G as its centroid. Let P be a variable point on segment BC . Points Q and R lie on sides AC and AB respectively, such that $PQ \parallel AB$ and $PR \parallel AC$. Prove that, as P varies along segment BC , the circumcircle of triangle AQR passes through a fixed point X such that $\angle BAG = \angle CAX$.

Problem 19. Let ABC be an acute triangle. Denote by D the foot of the perpendicular line drawn from the point A to the side BC , by M the midpoint of BC , and by H the orthocentre of $\triangle ABC$. Let E be the point of intersection of the circumcircle Γ of $\triangle ABC$ and the half line MH , and F be the point of intersection, other than E , of the line ED and the circle Γ . Prove that $\frac{BF}{CF} = \frac{AB}{AC}$ must hold.

Problem 20. In the triangle ABC let B' and C' be the midpoints of the sides AC and AB respectively and H the foot of the altitude passing through the vertex A . Prove that the circumcircles of the triangles $AB'C'$, $BC'H$, and $B'CH$ have a common point I and that the line HI passes through the midpoint of the segment $B'C'$.

Problem 21. Let ABC be an acute-angled triangle with $AB \neq AC$, circumcentre O and circumcircle Γ . Let the tangents to Γ at B and C meet each other at D , and let the line AO intersect BC at E . Denote the midpoint of BC by M and let AM meet Γ again at $N \neq A$. Finally, let $F \neq A$ be a point on Γ such that A, M, E and F are concyclic. Prove that FN bisects the segment MD .

Problem 22. Let Γ be the circumcircle of an acute $\triangle ABC$. Let M be the midpoint of BC , and let D be the midpoint of the minor arc BC of Γ . Let Ω be the circle with centre D that passes through M . The two tangents from A to Ω meet the side BC at E and F respectively, with E lying between B and F . A circle ω_1 is tangent to the segments AE and CE , and is also tangent to Γ at P . A circle ω_2 is tangent to the segments AF and BF , and is also tangent to Γ at Q . Let X be the intersection point of EP and FQ . Prove that $\angle EAX = \angle MAF$.